

[illegible]



```

kali@kali:~$ sudo sysctl -w net.ipv6.conf.all.use_tempaddr=2
sysctl -w net.ipv6.conf.all.use_tempaddr=2: success
kali@kali:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1::1 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 fd00::521:68be:8c4d:f667/64 scope link noprefixroute
        valid_lft 81765sec preferred_lft 81765sec

```

3. Temporary IPv6 Address

Sudo sysctl -w net.ipv6.conf.all.use_tempaddr=

Check Addresses

ip -6 addr show

```

kali@kali:~$ sudo sysctl -w net.ipv6.conf.all.use_tempaddr=2
sysctl -w net.ipv6.conf.all.use_tempaddr=2: success
kali@kali:~$ ip -6 addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 state UNKNOWN qlen 1000
    inet6 ::1::1 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 state UP qlen 1000
    inet6 fd00::521:68be:8c4d:f667/64 scope link noprefixroute
        valid_lft 81765sec preferred_lft 81765sec

```

4. Restaurant Search Script (Python)

```

restaurants = {
    "Pizza Hut": "fd00:1::10",
    "Dominos": "fd00:1::20",
    "Burger King": "fd00:1::30"
}

```



```
search = input("Enter restaurant name: ").lower()
```

```
for name, ipv6 in restaurants.items():
```

```
    if search in name.lower():
```

```
        print(name, "->", ipv6)
```

Example Input

pizza

Output

Pizza Hut -> fd00:1::10

5. AI-Generated IPv6 Subnetting Scenario

Configuration Command

```
sudo ip -6 addr add 2001:db8:1000:1::50/64 dev eth0
```

verification

```
ip -6 addr show eth0
```